

Machine Learning Analysis Report

Overview

This project addressed a supervised machine learning classification problem focused on predicting airline passenger satisfaction using the **Airline Passenger Satisfaction** dataset obtained from Kaggle. The dataset contains passenger demographic information, travel characteristics, service quality ratings, and flight delay information that were used to classify passengers as either *satisfied* or *neutral/dissatisfied*. Multiple machine learning approaches were evaluated, including Logistic Regression and Random Forest models, with a Random Forest classifier ultimately selected as the final model due to its superior predictive performance and ability to capture complex nonlinear relationships within the data.

Dataset Source: Kaggle

Dataset Link: <https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction>

Github Repo Link: <https://github.com/Shehzaad-S/udacity-applied-ml-project>

Dataset Description

The Airline Passenger Satisfaction dataset contains passenger survey responses and travel-related information collected from commercial airline customers. The dataset includes 103,904 training observations, 25,976 testing observations, and 25 predictor variables describing passenger demographics, travel characteristics, flight delays, and service quality ratings across various aspects of the airline experience.

The feature set includes variables such as passenger age, customer type, travel class, type of travel, flight distance, departure and arrival delays, and multiple service-related ratings including inflight Wi-Fi service, online boarding, seat comfort, baggage handling, and inflight entertainment. The target variable, **satisfaction**, is a binary classification outcome that categorizes passengers as either satisfied or neutral/dissatisfied.

Initial inspection indicated that the dataset was generally clean and well-structured for machine learning applications. No duplicate records were identified, only a single feature (arrival delay) contained a small proportion of missing values, and the target variable was reasonably balanced across both classes. These characteristics made the dataset well suited for supervised classification modeling while requiring only modest preprocessing prior to model development.

Modeling Approach

Data Preparation

The dataset required relatively modest preprocessing prior to model development. Two identifier variables (Unnamed: 0 and id) were removed because they did not contain meaningful predictive information. Missing values were present only within the Arrival Delay in Minutes feature and represented less than 1% of the training data. These missing observations were imputed using the median value within a preprocessing pipeline to preserve all available records while reducing the influence of extreme delay values.

Categorical variables, including gender, customer type, travel type, and travel class, were encoded using one-hot encoding. Service-related survey responses were retained as ordinal numerical variables because their underlying rating scales contain meaningful ordering information. Continuous variables, including age, flight distance, and delay measures, were initially retained in their original form. Additional experiments incorporating feature scaling and logarithmic transformations of delay variables were later conducted to evaluate their impact on model performance.

Model Development and Selection

Two machine learning algorithms were evaluated during model development: Logistic Regression and Random Forest. Logistic Regression was selected as an interpretable baseline model for binary classification, Random Forest was selected because, as established by Breiman (2001), its ensemble mechanism successfully handles complex, low-information feature sets and output noise by balancing individual tree strength with low mutual correlation.

Rather than assuming one algorithm would outperform the other, both models were trained and evaluated using the same training and testing datasets. Additional experiments were conducted using feature scaling and logarithmic transformations of the highly skewed delay variables to determine whether preprocessing improvements could enhance model performance. This approach aligns with recent methodology emphasizing that performance in tabular data classification is heavily dictated by data characteristics and quality rather than model complexity alone, shifting the focus from algorithmic superiority debates toward data quality and modeling procedures (Hu et al., 2025).

Although feature scaling substantially improved Logistic Regression convergence and log transformations reduced skewness within the delay variables, predictive performance improved only marginally. Random Forest consistently achieved superior performance

across all evaluation metrics and was therefore selected as the final model. Hyperparameter tuning was subsequently performed to determine whether additional performance improvements could be achieved. The tuned Random Forest model produced performance nearly identical to the baseline model, indicating that the original configuration was already highly effective.

Evaluation Metrics

Model performance was assessed using Accuracy, Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic curve (ROC-AUC). Evaluating multiple metrics provides a comprehensive assessment of classification performance because different metrics capture distinct aspects of predictive behavior, preventing over-reliance on a single measure (Fawcett, 2006).

Accuracy was used to measure overall classification performance, while Precision and Recall were evaluated to determine the model's ability to correctly identify satisfied versus neutral or dissatisfied passengers. The F1-score was included to provide a balanced harmonic mean of Precision and Recall. Finally, ROC-AUC was used to assess the model's ability to distinguish between satisfaction classes across a range of classification thresholds.

Model Assumptions

The baseline Logistic Regression model assumes a linear relationship between predictor variables and the log-odds of the target outcome. Logistic Regression also assumes that observations are independent and that excessive multicollinearity is not present among predictors. Violations of these assumptions can reduce coefficient stability, limit interpretability, and negatively affect predictive performance.

Random Forest makes substantially fewer assumptions about the underlying data generating process. Unlike Logistic Regression, Random Forest does not require linear relationships between predictors and the target variable, does not assume normally distributed variables, and is generally more robust to multicollinearity and complex interactions among features through its ensemble tree-based structure (Breiman, 2001). These characteristics make Random Forest particularly suitable for datasets containing nonlinear relationships, mixed variable types, and interaction effects.

At the dataset level, both modeling approaches fundamentally assume that individual observations are independent and that the training and testing sets are representative of the same underlying passenger population. The analysis also assumes that passenger demographics, travel characteristics, service ratings, and flight delay information collectively contain sufficient information to predict customer satisfaction. As noted by Hu

et al. (2025), model performance is influenced not only by algorithm selection but also by dataset characteristics, data quality, preprocessing decisions, feature engineering choices, and the overall modeling procedure. The large sample size and relatively balanced class distribution of the Airline Passenger Satisfaction dataset helped support stable model training and evaluation.

Results

The initial model comparison showed a substantial performance difference between Logistic Regression and Random Forest. The baseline Logistic Regression model achieved an accuracy of **87.1%**, an F1-score of **0.8499**, and a ROC-AUC score of **0.9257**, while the Random Forest model achieved **96.3% accuracy**, an **F1-score of 0.9573**, and a **ROC-AUC score of 0.9943**. Additional preprocessing experiments involving feature scaling and logarithmic transformation of the delay variables improved Logistic Regression convergence but produced only marginal improvements in predictive performance. As shown in **Figure 1 (see below)**, Random Forest consistently outperformed all Logistic Regression variants across every evaluation metric and was therefore selected as the final model.

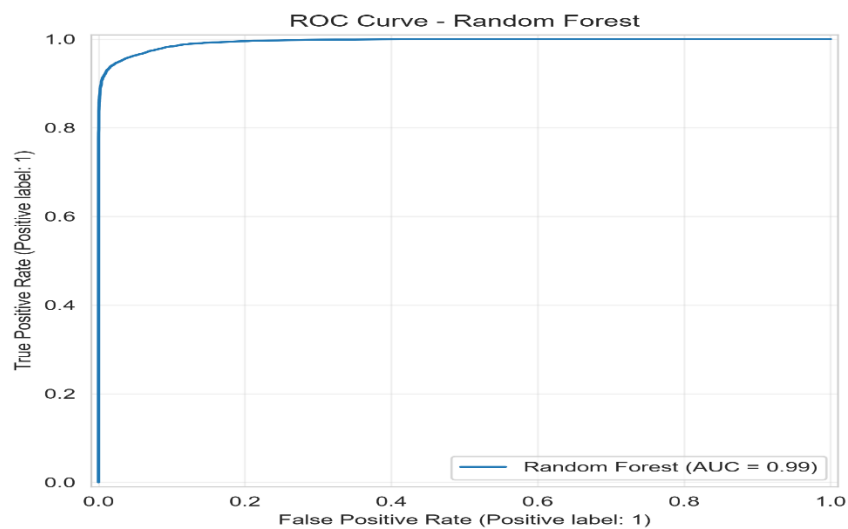
Figure 1: Model Performance Comparison

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.8706	0.8658	0.8345	0.8499	0.9257
1	Scaled + Log Logistic Regression	0.8717	0.8692	0.8331	0.8508	0.9270
2	Random Forest	0.9630	0.9721	0.9428	0.9573	0.9943

Hyperparameter tuning was conducted on the Random Forest model to determine whether additional performance improvements could be achieved. Although the optimal parameter combination increased model complexity through a larger number of trees and deeper tree structures, performance remained nearly identical to the baseline model. This finding suggested that the original Random Forest configuration was already highly effective and did not require additional tuning to achieve strong predictive performance.

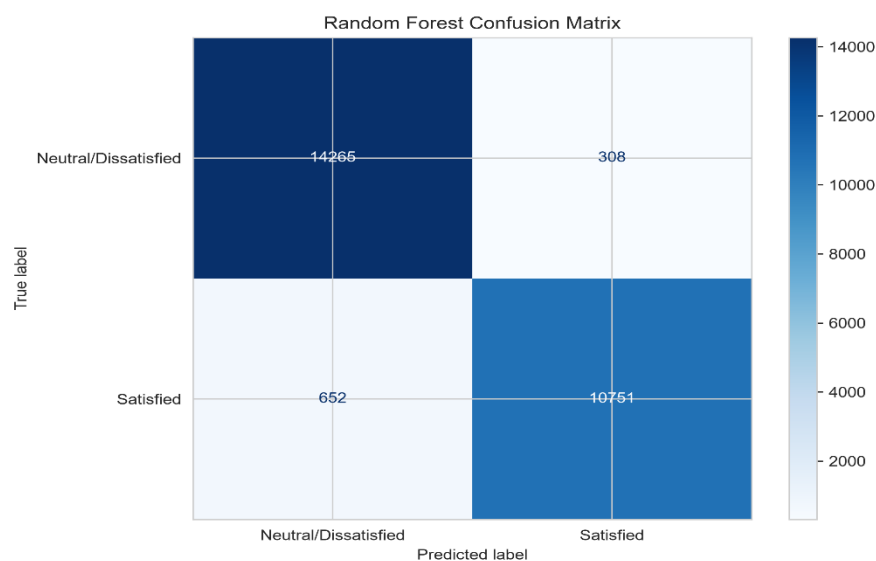
These results indicated that the model was highly effective at distinguishing between satisfied passengers and those who were neutral or dissatisfied. The ROC curve shown in **Figure 2** remained close to the upper-left corner of the plot, indicating excellent class separation across a wide range of classification thresholds.

Figure 2: ROC Curve for Final Random Forest Model



The confusion matrix presented in **Figure 3** provides additional insight into model performance. The model correctly classified **14,265 neutral or dissatisfied passengers** and **10,751 satisfied passengers**, while producing only **960 total misclassifications** across the testing dataset. The model generated relatively few false positive predictions, indicating that passengers predicted as satisfied were usually classified correctly. A slightly larger number of false negatives were observed, suggesting that the model was somewhat conservative when assigning the satisfied class.

Figure 3: Random Forest Confusion Matrix



The classification report further confirmed strong and consistent performance across both target classes. Precision, recall, and F1-scores exceeded **0.94** for both satisfied and neutral or dissatisfied passengers, demonstrating that the model did not heavily favor one class over the other despite the slight class imbalance present in the dataset.

To improve model interpretability, SHAP analysis was performed on the final Random Forest model. The SHAP summary plot shown in **Figure 4** identified **Inflight WiFi Service** as the most influential predictor of passenger satisfaction, followed by travel purpose, online boarding experience, customer loyalty status, and several airline service quality measures. Passengers with higher Wi-Fi ratings and stronger digital service experiences were more likely to be classified as satisfied, while personal travel and disloyal customer status tended to push predictions toward the neutral or dissatisfied class. These findings indicate that passenger satisfaction is influenced by a combination of digital convenience, loyalty characteristics, travel context, and overall service quality rather than any single factor alone.

Figure 4: SHAP Summary Plot



Interpretation for a Non-Technical Audience

The final Random Forest model was highly effective at predicting whether an airline passenger would be satisfied with their travel experience. In practical terms, the model

correctly classified passenger satisfaction outcomes in approximately **96% of cases**, indicating that it was able to identify patterns associated with customer satisfaction with a high degree of accuracy.

The analysis suggests that passenger satisfaction is influenced by a combination of factors rather than a single aspect of travel experience. Features such as **Inflight WiFi Service**, **travel purpose**, **customer loyalty status**, **online boarding experience**, and other service quality measures played important roles in the model's predictions. Passengers who reported stronger digital service experiences and higher service quality ratings were generally more likely to be classified as satisfied.

From a business perspective, these findings suggest that improvements to customer-facing digital services and overall service quality may have a meaningful impact on passenger satisfaction.

Limitations and Potential Bias

Several limitations should be considered when interpreting these findings. First, the dataset represents a specific airline passenger population and may not fully generalize to other airlines, geographic regions, customer segments, or future travel environments.

A second limitation relates to the available features. While the dataset contains extensive information on service quality, travel characteristics, and delays, it does not include potentially relevant factors such as ticket price, airline-specific policies, weather conditions, flight disruptions, or individual passenger expectations. These omitted variables may influence satisfaction outcomes and could explain some of the remaining prediction errors.

Potential sources of bias and uncertainty were also identified during the explainability analysis. Features such as **customer loyalty status** and **travel purpose** emerged as some of the most influential predictors of satisfaction. As a result, the model may partially reflect behavioral patterns specific to loyal customers or business travelers rather than purely measuring service quality. This does not necessarily indicate unfairness, but it highlights the possibility that different passenger groups may be represented differently within the model's decision-making process.

Finally, although SHAP analysis improved model transparency by identifying the primary drivers of satisfaction predictions, a formal fairness assessment was not conducted. One proposed mitigation strategy would be to evaluate model performance across demographic and customer subgroups using a dedicated fairness assessment framework such as Aequitas. This would allow the identification of potential disparities in prediction outcomes and support future model refinements if systematic differences between

passenger groups were detected. Potential mitigation approaches could include feature review, decision-threshold adjustment, or reweighing techniques that reduce the influence of group imbalances. However, a **critical consideration** for future expansion is the trade-off inherent in data-level interventions. While preprocessing techniques such as reweighing can mitigate disparate impact, they also modify the original data distribution and may affect model interpretability, predictive performance, or generalization to unseen data. Consequently, future iterations should carefully evaluate these trade-offs against alternative in-processing or post-processing approaches that seek to improve fairness while preserving as much of the original data structure as possible.

References

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